1) $\int \frac{\sqrt{x+1}+1}{\sqrt{x+1}-1}dx = \left| t=\sqrt{x+1} \Rightarrow x=t^2-1 \Rightarrow dx=2tdt \right| = \int \frac{t+1}{t-1}\cdot 2tdt = \int \left(1+\frac{2}{t-1}\right)2tdt =$

$= \int 2tdt + 4\int \underbrace{\frac{t}{t-1}}_{1+\frac{1}{t-1}}dt = t^2 + 4\int\left(1+\frac{1}{t-1}\right)dt = t^2+4t+4\ln|t-1|+C =$

$= x+1+4\sqrt{x+1}+4\ln\left|\sqrt{x+1}-1\right|+C =$

$= x+4\sqrt{x+1}+4\ln\left|\sqrt{x+1}-1\right|+C.$

2) $\int \frac{x^3-6x^2+11x-6}{\sqrt{x^2+4x+3}}dx = Q_2(x)\cdot\sqrt{x^2+4x+3} + \int \frac{\lambda dx}{\sqrt{x^2+4x+3}}$ (=)

$Q_2(x) = Ax^2+Bx+C$

$\frac{x^3-6x^2+11x-6}{\sqrt{x^2+4x+3}} = (2Ax+B)\sqrt{x^2+4x+3} + \frac{(Ax^2+Bx+C)(x+2)}{\sqrt{x^2+4x+3}} + \frac{\lambda}{\sqrt{x^2+4x+3}} \Rightarrow$

$\Rightarrow x^3-6x^2+11x-6 = (2Ax+B)(x^2+4x+3)+(Ax^2+Bx+C)(x+2)+\lambda$

$x^3-6x^2+11x-6 = 2Ax^3+Bx^2+8Ax^2+4Bx+6Ax+3B+Ax^3+Bx^2+Cx+2Ax^2+2Bx+2C+\lambda$

$1=3A \Rightarrow A=\frac{1}{3}$

$-6 = B+10A+B \Rightarrow -6 = 2B+\frac{10}{3} \Rightarrow -18 = 6B+10 \Rightarrow B = -\frac{28}{6} = -\frac{14}{3}$

$11 = 4B+6A+C+2B \Rightarrow 11 = 6B+6A+C \Rightarrow 11 = -28+2+C \Rightarrow C = 37$

$-6 = 3B+2C+\lambda \Rightarrow -6 = -14+74+\lambda \Rightarrow \lambda = -66$

(=) $\left(\frac{x^2}{3}-\frac{14}{3}x+37\right)\sqrt{x^2+4x+3} - 66\int\frac{dx}{\sqrt{(x+2)^2-1}} = \frac{(x^2-14x+111)\sqrt{x^2+4x+3}}{3} - 66\ln\left|x+2+\sqrt{x^2+4x+3}\right|+C$

3) $\int \frac{dx}{(x+1)\sqrt{x^2+x+1}} = \left| \begin{array}{l} t=\frac{1}{x+1} \Rightarrow x=\frac{1}{t}-1 \Rightarrow dx=-\frac{dt}{t^2} \\ x+1=\frac{1}{t} \\ x^2=\frac{1}{t^2}-\frac{2}{t}+1 \\ x^2+x+1=\frac{1}{t^2}-\frac{1}{t}+1=\frac{t^2-t+1}{t^2} \end{array} \right| = -\int\frac{dt}{t}\cdot\frac{1}{\frac{\sqrt{t^2-t+1}}{t}} = -\int\frac{dt}{\sqrt{(t-0{,}5)^2+\frac{3}{4}}} =$

$= -\ln\left|t-\frac{1}{2}+\sqrt{t^2-t+1}\right|+C$

4) $\int \frac{dx}{(x^2+1)^{\frac{x}{2}}} = \int \frac{dx}{(x^2+1)\sqrt[3]{x^2+1}} = \left| \begin{array}{l} t=\left(\sqrt{x^2+1}\right)' = \frac{x}{\sqrt{x^2+1}} \\ t\sqrt{x^2+1}=x \\ dx = dt\sqrt{x^2+1}+t^2dx \Rightarrow dx = \frac{dt\sqrt{x^2+1}}{1-t^2} \\ t^2 = \frac{x^2}{x^2+1} = 1-\frac{1}{x^2+1} \Rightarrow x^2+1 = \frac{1}{1-t^2} \end{array} \right| =$

$= \int \frac{dt\sqrt{x^2+1}}{1-t^2}\cdot(1-t^2)^{\frac{3}{2}}\cdot\frac{1}{\sqrt{x^2+1}} = \int(1-2t^2+t^4)dt = t-\frac{2}{3}t^3+\frac{t^5}{5}+C$

5) $\int \frac{1+\sqrt{1+x+x^2}}{1+x+\sqrt{1+x+x^2}}dx = \int\left(\sqrt{1+x+x^2}-x\right)dx = -\frac{x^2}{2}+\int\sqrt{1+x+x^2}dx$ (=)

$\cdot\frac{1+\sqrt{1+x+x^2}}{(1+x)+\sqrt{1+x+x^2}} = \frac{\left(1+\sqrt{1+x+x^2}\right)\left(1+x-\sqrt{1+x+x^2}\right)}{x} = \frac{1+\sqrt{1+x+x^2}+x+x\sqrt{1+x+x^2}-\sqrt{1+x+x^2}-1-x-x^2}{x} =$

$= \frac{x\sqrt{1+x+x^2}-x^2}{x}$

$= \sqrt{1+x+x^2} - x$

$\cdot \int \sqrt{1+x+x^2}\,dx = \left| \begin{matrix} u = \sqrt{1+x+x^2}, & du = \frac{2x+1}{2\sqrt{1+x+x^2}} \\ v = x \end{matrix} \right| = x\sqrt{1+x+x^2} - \frac{1}{2}\int \frac{x(2x+1)}{1+x+x^2}\sqrt{1+x+x^2}\,dx =$

$\cdot -\frac{1}{2}\int \frac{2x^2+x}{x^2+x+1}\sqrt{x^2+x+1}\,dx = -\frac{1}{2}\int \left(2 - \frac{x-2}{x^2+x+1}\right)\sqrt{x^2+x+1}\,dx =$

$= -\int \sqrt{x^2+x+1}\,dx + \frac{1}{2}\int \frac{x-2}{\sqrt{x^2+x+1}}\,dx = -\int \sqrt{x^2+x+1}\,dx + \frac{1}{4}\int \frac{(2x+1)\,dx}{\sqrt{x^2+x+1}} - \frac{5}{4}\int \frac{dx}{\sqrt{(x+\frac{1}{2})^2+\frac{3}{4}}} =$

$= -\int \sqrt{x^2+x+1}\,dx + \frac{\sqrt{x^2+x+1}}{2} - \frac{5}{4}\ln\left|x+0{,}5+\sqrt{x^2+x+1}\right| + C$

$= x\sqrt{x^2+x+1} - \int \sqrt{x^2+x+1}\,dx + \frac{\sqrt{x^2+x+1}}{2} - \frac{5}{4}\ln\left|x+0{,}5+\sqrt{x^2+x+1}\right| + C \Rightarrow$

$\Rightarrow \int \sqrt{x^2+x+1} = \frac{x\sqrt{x^2+x+1}}{2} + \frac{\sqrt{x^2+x+1}}{4} - \frac{5}{8}\ln\left|x+0{,}5+\sqrt{x^2+x+1}\right| + C$

$= -\frac{x^2}{2} + \frac{x\sqrt{x^2+x+1}}{2} + \frac{\sqrt{x^2+x+1}}{4} - \frac{5}{8}\ln\left|x+\frac{1}{2}+\sqrt{x^2+x+1}\right| + C$

6) $\int \frac{dx}{x\sqrt[6]{x^6+1}} = \left| \begin{matrix} t^6 = x^6+1 \\ x = \sqrt[6]{t^6-1} \\ dx = \frac{t^5\,dt}{\sqrt[6]{(t^6-1)^5}} \end{matrix} \right| = \int \frac{t^5 t^4}{\sqrt[6]{(t^6-1)^5}\cdot \sqrt[6]{t^6-1}\,t} = \int \frac{t^4\,dt}{t^6-1} = \int \frac{t^4\,dt}{(t^2-1)(t^4+t^2+1)} =$

$= \int \frac{(t^4+t^2+1)-(t^2+1)}{(t^2-1)(t^4+t^2+1)}\,dt = \int \frac{dt}{t^2-1} - \int \frac{t^2\,dt}{(t^3)^2-1} - \int \frac{dt}{t^6-1} = \frac{1}{2}\ln\left|\frac{t-1}{t+1}\right| - \frac{1}{6}\ln\left|\frac{t^3-1}{t^3+1}\right| -$

($\frac{1}{3}dt^3$; $\frac{1}{2}\ln\left|\frac{t-1}{t+1}\right|$; $-\frac{1}{6}\ln\left|\frac{t^3-1}{t^3+1}\right|$)

$-\int \frac{dt}{t^6-1} =$

$\Delta\ -\int \frac{dt}{t^6-1} = -\int \frac{t^2-(t^2-1)}{t^6-1}\,dt = -\int \frac{t^2\,dt}{t^6-1} + \int \frac{dt}{t^4+t^2+1} = -\frac{1}{6}\ln\left|\frac{t^3-1}{t^3+1}\right| + \frac{1}{2}\int \frac{(t^2+1)-(t^2-1)}{t^4+t^2+1}\,dt =$

$= -\frac{1}{6}\ln\left|\frac{t^3-1}{t^3+1}\right| + \frac{1}{2}\int \frac{1+\frac{1}{t^2}}{t^2+1+\frac{1}{t^2}}\,dt - \frac{1}{2}\int \frac{1-\frac{1}{t^2}}{t^2+1+\frac{1}{t^2}}\,dt = -\frac{1}{6}\ln\left|\frac{t^3-1}{t^3+1}\right| + \frac{1}{2}\int \frac{d\left(t-\frac{1}{t}\right)}{\left(t-\frac{1}{t}\right)^2+3} -$

$-\frac{1}{2}\int \frac{d\left(t+\frac{1}{t}\right)}{\left(t+\frac{1}{t}\right)^2-1} =$

$= -\frac{1}{6}\ln\left|\frac{t^3-1}{t^3+1}\right| + \frac{1}{2\sqrt{3}}\operatorname{arctg}\left(\frac{t-\frac{1}{t}}{\sqrt{3}}\right) - \frac{1}{4}\ln\left|\frac{t+\frac{1}{t}-1}{t+\frac{1}{t}+1}\right| + C$

$= \frac{1}{2}\ln\left|\frac{t-1}{t+1}\right| - \frac{1}{3}\ln\left|\frac{t^3-1}{t^3+1}\right| + \frac{1}{2\sqrt{3}}\operatorname{arctg}\left(\frac{t^2-1}{\sqrt{3}t}\right) - \frac{1}{4}\ln\left|\frac{t^2-t+1}{t^2+t+1}\right| + C$